

Linux Commands with Examples for DevOps Labs:

• File & Directory Management:

1. Command: pwd

Purpose: Show current directory

Example & Result:

```
student@Riya:~$ pwd  
/home/student
```

2. Command: ls

Purpose: List files

Example:

```
student@Riya:~$ ls  
Desktop  Downloads  Music      Public    Student   Videos  
Documents j.zip      Pictures   snap      Templates
```

3. Command: mkdir

Purpose: Create directory

Example:

```
student@Riya:~$ mkdir Radha
```

Result:



4. Command: cd

Purpose: Change directory

Example & Result:

```
student@Riya:~$ cd Radha
```

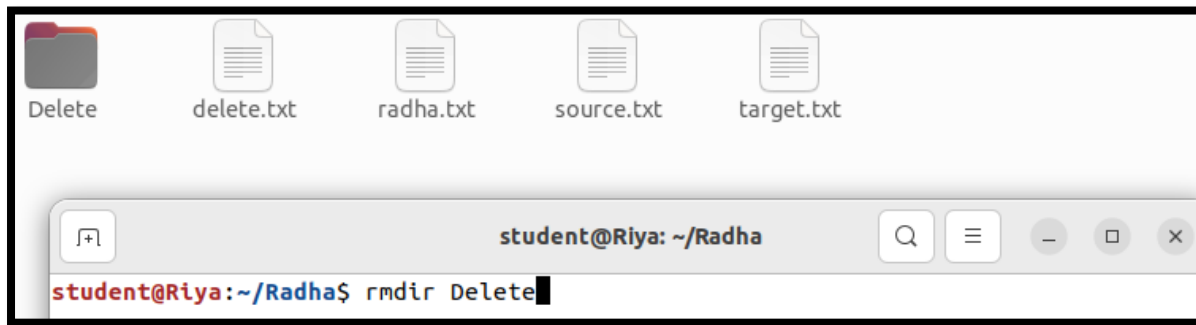
5. Command: touch

Purpose: Creates an empty file.

Example:

```
student@Riya:~/Radha$ touch radha.txt source.txt target.txt delete.txt  
student@Riya:~/Radha$
```

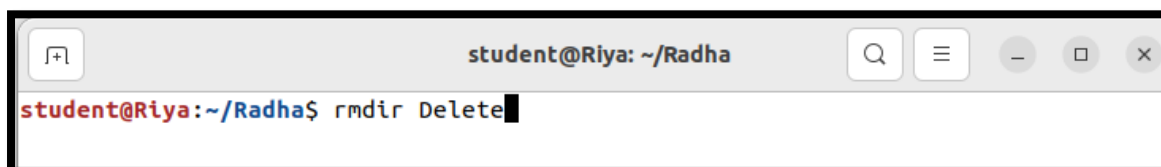
Result:



6. Command: rmdir

Purpose: Deletes empty directory

Example:



Result:

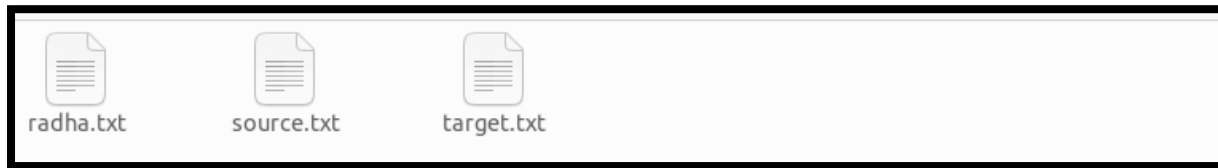
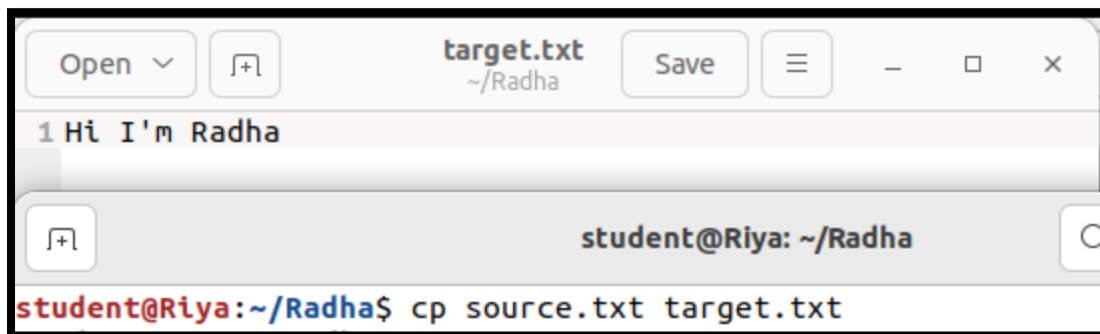
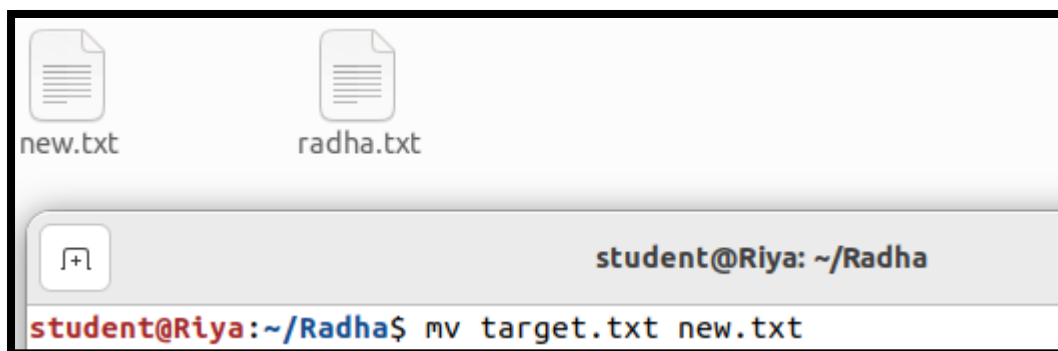


7. Command: rm

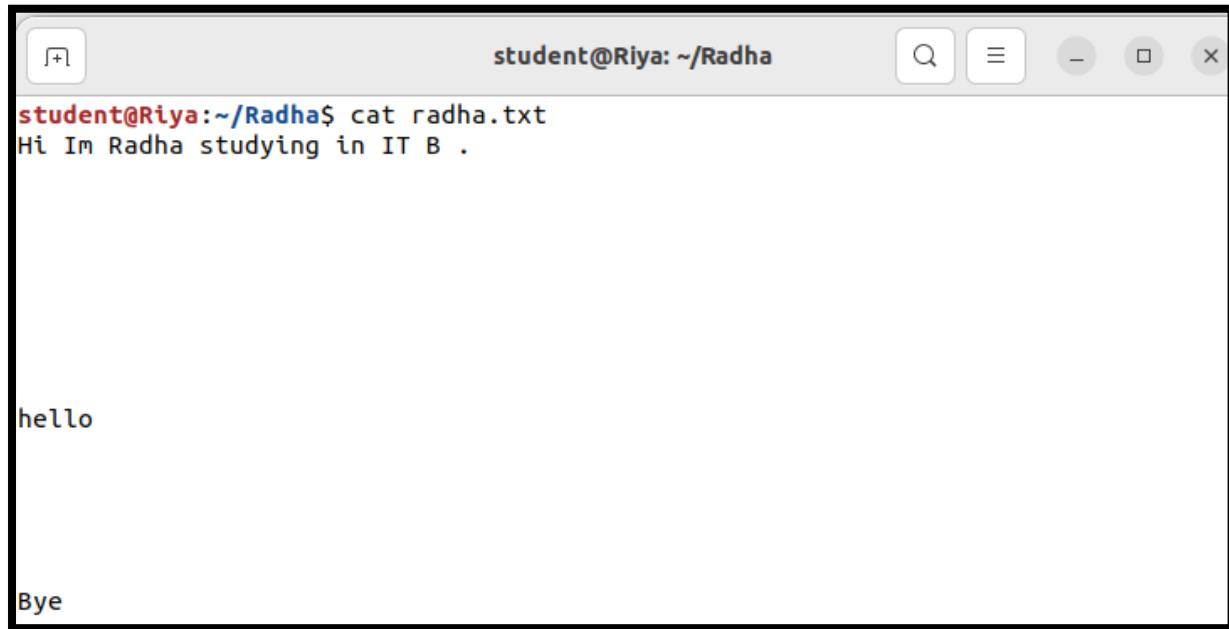
Purpose: Delete file/folder

Example:

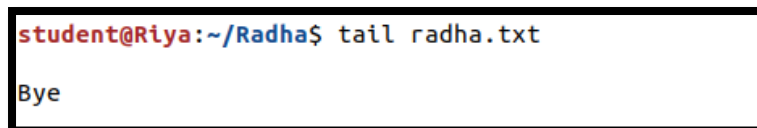
```
student@Riya:~/Radha$ rm delete.txt  
student@Riya:~/Radha$
```

Result:**8. Command: cp****Purpose:** Copy files**Example & Result:****9. Command: mv****Purpose:** Move/Rename files**Example & Result:**

- **File Viewing & Editing**

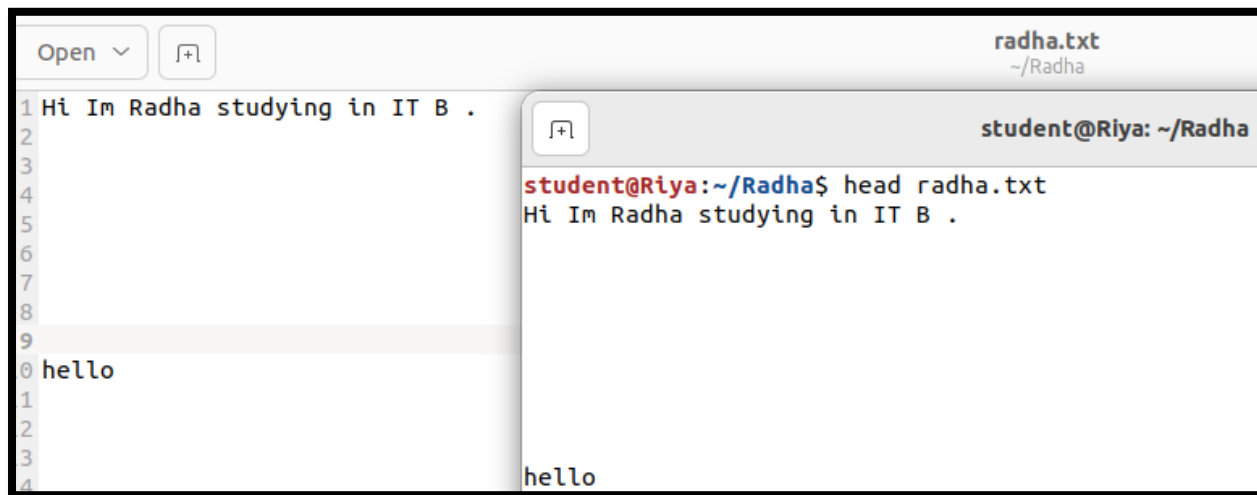
10. Command: cat**Purpose:** Display file content**Example & Result:**A terminal window titled 'student@Riya: ~/Radha' with search, menu, and window control icons. The prompt is 'student@Riya:~/Radha\$'. The command 'cat radha.txt' has been entered, and the output is displayed line by line: 'Hi Im Radha studying in IT B .', 'hello', and 'Bye'.

```
student@Riya:~/Radha$ cat radha.txt
Hi Im Radha studying in IT B .
hello
Bye
```

11. Command: tail**Purpose:** Display last lines of file**Example & Result:**A terminal window showing the prompt 'student@Riya:~/Radha\$' and the command 'tail radha.txt'. The output 'Bye' is visible.

```
student@Riya:~/Radha$ tail radha.txt
Bye
```

12. Command: head**Purpose:** Display first lines of file**Example & Result:**



The screenshot shows a terminal window with two panes. The left pane shows the nano text editor editing a file named 'radha.txt' in the directory '~/Radha'. The file content is: 'Hi Im Radha studying in IT B .' on line 1, and 'hello' on line 9. The right pane shows a terminal window with the prompt 'student@Riya: ~/Radha'. The command 'head radha.txt' has been executed, and the output is 'Hi Im Radha studying in IT B .'.

13. Command: nano

Purpose: Edit file

Example:

```
student@Riya:~$ nano radha.txt
```

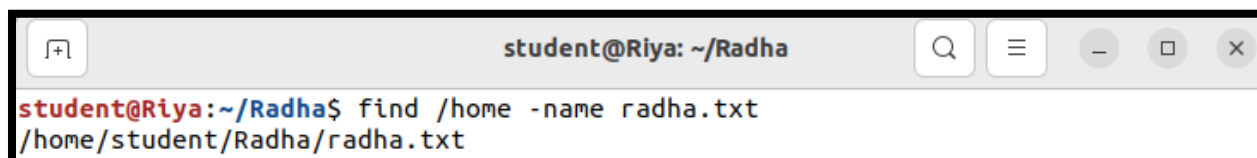
Result : Opens text editor

• Search & System Information

14. Command: find

Purpose: Find files

Example & Result:



The screenshot shows a terminal window with the prompt 'student@Riya: ~/Radha'. The command 'find /home -name radha.txt' has been executed, and the output is '/home/student/Radha/radha.txt'.

15. Command: grep

Purpose: Search text

Example & Result:

```
student@Riya:~/Radha$ grep 'hello' radha.txt
hello
```

16. Command: whoami**Purpose:** Current user**Example & Result:**

```
student@Riya:~/Radha$ whoami  
student
```

17. Command: uname**Purpose:** System info**Example & Result:**

```
student@Riya:~/Radha$ uname -a  
Linux Riya 6.8.0-65-generic #68~22.04.1-Ubuntu SMP PREEMPT_DYNAMIC Tue Jul 15 18  
:06:34 UTC 2 x86_64 x86_64 x86_64 GNU/Linux  
student@Riya:~/Radha$
```

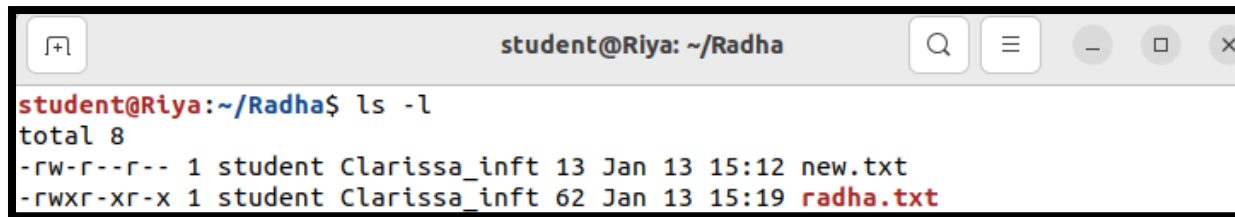
- **Permissions & Ownership**

18. Command: chmod**Purpose:** Change permissions**Example & Result:**

```
student@Riya: ~/Radha  
student@Riya:~/Radha$ chmod 755 radha.txt
```

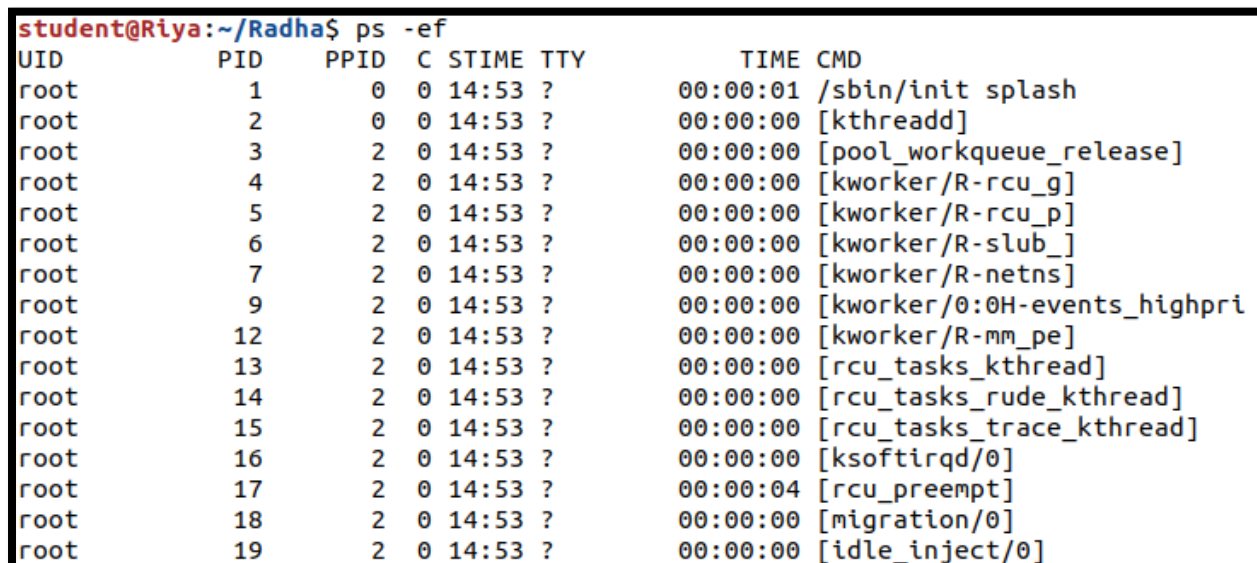
19. Command: chown**Purpose:** Change owner**Example & Result:**

```
student@Riya:~/Radha$ sudo chown user:user radha.txt  
[sudo] password for student:
```

20. Command: `ls -l`**Purpose:** View permissions**Example & Result:**

```
student@Riya: ~/Radha
student@Riya:~/Radha$ ls -l
total 8
-rw-r--r-- 1 student Clarissa_inft 13 Jan 13 15:12 new.txt
-rwxr-xr-x 1 student Clarissa_inft 62 Jan 13 15:19 radha.txt
```

- **Process & System Management**

21. Command: `ps`**Purpose:** Running processes**Example & Result:**

```
student@Riya:~/Radha$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1         0  0 14:53 ?        00:00:01 /sbin/init splash
root           2         0  0 14:53 ?        00:00:00 [kthreadd]
root           3         2  0 14:53 ?        00:00:00 [pool_workqueue_release]
root           4         2  0 14:53 ?        00:00:00 [kworker/R-rcu_g]
root           5         2  0 14:53 ?        00:00:00 [kworker/R-rcu_p]
root           6         2  0 14:53 ?        00:00:00 [kworker/R-slub_]
root           7         2  0 14:53 ?        00:00:00 [kworker/R-netns]
root           9         2  0 14:53 ?        00:00:00 [kworker/0:0H-events_highpri]
root          12         2  0 14:53 ?        00:00:00 [kworker/R-mm_pe]
root          13         2  0 14:53 ?        00:00:00 [rcu_tasks_kthread]
root          14         2  0 14:53 ?        00:00:00 [rcu_tasks_rude_kthread]
root          15         2  0 14:53 ?        00:00:00 [rcu_tasks_trace_kthread]
root          16         2  0 14:53 ?        00:00:00 [ksoftirqd/0]
root          17         2  0 14:53 ?        00:00:04 [rcu_preempt]
root          18         2  0 14:53 ?        00:00:00 [migration/0]
root          19         2  0 14:53 ?        00:00:00 [idle inject/0]
```

22. Command: `top`**Purpose:** Real-time processes**Example & Result:**

```

student@Riya: ~/Radha
top - 15:28:26 up 34 min, 1 user, load average: 0.41, 0.43, 0.46
Tasks: 318 total, 2 running, 316 sleeping, 0 stopped, 0 zombie
%Cpu(s): 4.4 us, 0.4 sy, 0.0 ni, 95.0 id, 0.1 wa, 0.0 hi, 0.1 si, 0.0 st
MiB Mem : 15701.4 total, 10436.1 free, 1938.6 used, 3326.7 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 13064.4 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM     TIME+ COMMAND
  637 avahi     20   0   14068   10112   3456 R   45.0   0.1    6:53.59 avahi-d+
 1749 student   20   0 5684128 288420 136636 S    6.3   1.8    1:36.72 gnome-s+
 4676 student   20   0 553204   51232 38996 S    4.6   0.3    0:07.35 gnome-t+
  133 root      20   0      0      0      0 I    0.3   0.0    0:02.70 kworker+
  654 root      20   0 261332   18092 15532 S    0.3   0.1    0:11.50 Network+
 3470 student   20   0 889348   96464 57060 S    0.3   0.6    0:14.08 nautilus
 4464 root       0 -20      0      0      0 I    0.3   0.0    0:02.14 kworker+
    1 root      20   0 168248   12720  8112 S    0.0   0.1    0:01.62 systemd
    2 root      20   0      0      0      0 S    0.0   0.0    0:00.00 kthreadd
    3 root      20   0      0      0      0 S    0.0   0.0    0:00.00 pool_wo+
    4 root       0 -20      0      0      0 I    0.0   0.0    0:00.00 kworker+
    5 root       0 -20      0      0      0 I    0.0   0.0    0:00.00 kworker+
    6 root       0 -20      0      0      0 I    0.0   0.0    0:00.00 kworker+
    7 root       0 -20      0      0      0 I    0.0   0.0    0:00.00 kworker+
    9 root       0 -20      0      0      0 I    0.0   0.0    0:00.00 kworker+
   12 root       0 -20      0      0      0 I    0.0   0.0    0:00.00 kworker+
   13 root      20   0      0      0      0 I    0.0   0.0    0:00.00 rcu_tas+

```

23. Command: kill

Purpose: Terminate process

Example & Result:

```

student@Riya: ~/Radha
student@Riya:~/Radha$ kill 1234
bash: kill: (1234) - No such process

```

24. Command: df

Purpose: Disk usage

Example & Result:

```

student@Riya:~/Radha$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            1.6G  2.3M  1.6G   1% /run
/dev/nvme0n1p5  144G   21G  116G  16% /
tmpfs            7.7G   0  7.7G   0% /dev/shm
tmpfs            5.0M  4.0K  5.0M   1% /run/lock
efivarfs        192K  161K   27K  86% /sys/firmware/efi/efivars
/dev/nvme0n1p1   96M   49M   48M  51% /boot/efi
tmpfs            1.6G  108K  1.6G   1% /run/user/1000

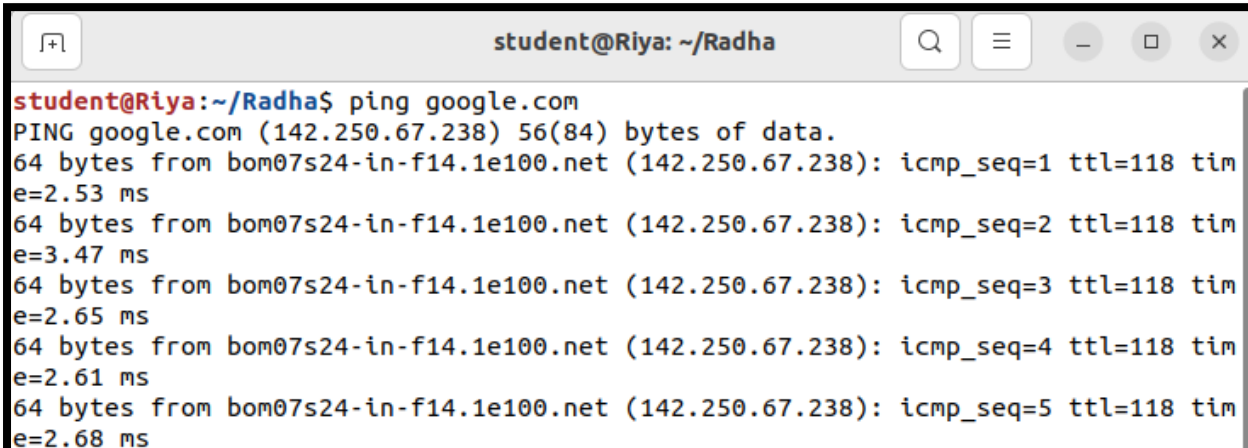
```


25. Command: free**Purpose:** Memory usage**Example & Result:**

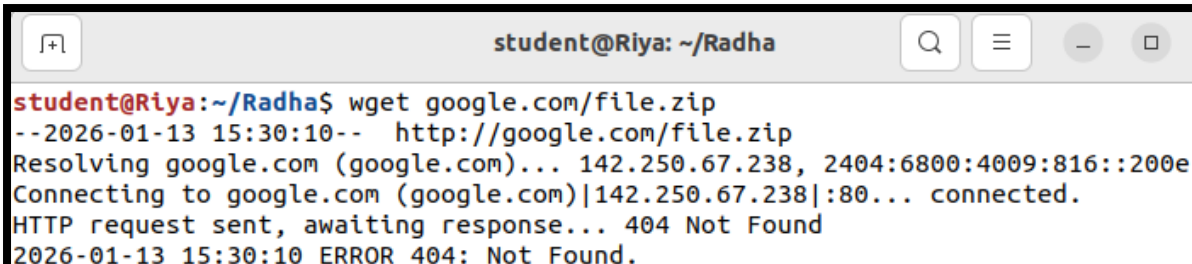
```
student@Riya:~/Radha$ free -m
```

	total	used	free	shared	buff/cache	available
Mem:	15701	1916	10467	665	3317	13097
Swap:	2047	0	2047			

- **Networking**

26. Command: ping**Purpose:** Test connectivity**Example & Result:**A terminal window titled 'student@Riya: ~/Radha' showing the output of the 'ping google.com' command. The output displays five successful ping attempts with varying response times.

```
student@Riya:~/Radha$ ping google.com
PING google.com (142.250.67.238) 56(84) bytes of data.
64 bytes from bom07s24-in-f14.1e100.net (142.250.67.238): icmp_seq=1 ttl=118 time=2.53 ms
64 bytes from bom07s24-in-f14.1e100.net (142.250.67.238): icmp_seq=2 ttl=118 time=3.47 ms
64 bytes from bom07s24-in-f14.1e100.net (142.250.67.238): icmp_seq=3 ttl=118 time=2.65 ms
64 bytes from bom07s24-in-f14.1e100.net (142.250.67.238): icmp_seq=4 ttl=118 time=2.61 ms
64 bytes from bom07s24-in-f14.1e100.net (142.250.67.238): icmp_seq=5 ttl=118 time=2.68 ms
```

27. Command: wget**Purpose:** Download file**Example & Result:**A terminal window titled 'student@Riya: ~/Radha' showing the output of the 'wget google.com/file.zip' command. The output indicates a 404 Not Found error.

```
student@Riya:~/Radha$ wget google.com/file.zip
--2026-01-13 15:30:10-- http://google.com/file.zip
Resolving google.com (google.com)... 142.250.67.238, 2404:6800:4009:816::200e
Connecting to google.com (google.com)|142.250.67.238|:80... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-01-13 15:30:10 ERROR 404: Not Found.
```

• Package Management (Ubuntu/Debian)

28. Command: apt update

Purpose: Update packages

Example & Result:

```
student@Riya:~/Radha$ sudo apt update
Get:1 https://dl.google.com/linux/chrome/deb stable InRelease [1,825 B]
Err:1 https://dl.google.com/linux/chrome/deb stable InRelease
  The following signatures couldn't be verified because the public key is not available: NO_PUBKEY 32EE5355A6BC6E42
Hit:2 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Get:3 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:4 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu jammy-updates/main i386 Packages [934 kB]
Get:7 http://security.ubuntu.com/ubuntu jammy-security/main i386 Packages [752 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [3,164 kB]
Get:9 http://in.archive.ubuntu.com/ubuntu jammy-updates/main Translation-en [485 kB]
```

29. Command: apt install

Purpose: Install software

Example & Result:

```
student@Riya: ~/Radha
student@Riya:~/Radha$ sudo apt install git
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  git-man liberror-perl
Suggested packages:
  git-daemon-run | git-daemon-sysvinit git-doc git-email git-gui gitk gitweb
  git-cvs git-mediawiki git-svn
The following NEW packages will be installed:
  git git-man liberror-perl
0 upgraded, 3 newly installed, 0 to remove and 182 not upgraded.
Need to get 4,147 kB of archives.
After this operation, 21.0 MB of additional disk space will be used.
Do you want to continue? [Y/n] n
Abort.
```

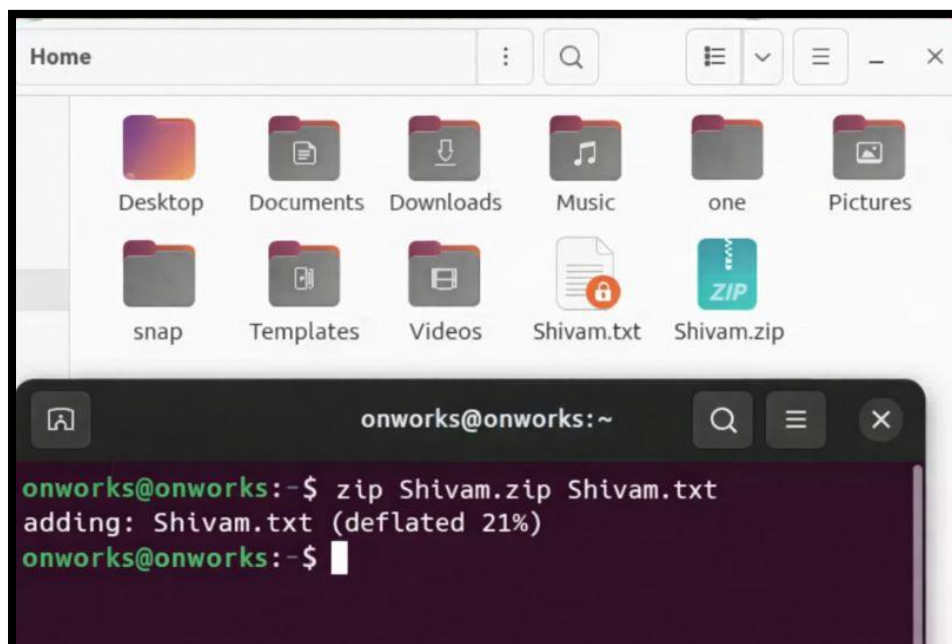
30. Command: apt remove**Purpose:** Remove software**Example & Result:**

```
student@Riya:~/Radha$ sudo apt remove git
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package 'git' is not installed, so not removed
0 upgraded, 0 newly installed, 0 to remove and 182 not upgraded.
```

- **Archive & Compression**

31. Command: tar**Purpose:** Create/Extract archive**Example & Result:**

```
student@Riya: ~/Radha
student@Riya:~/Radha$ tar -cvf data.tar data.txt
data.txt
```

32. Command: zip**Purpose:** Compress files**Example & Result:**

33. Command: unzip**Purpose:** Extract zip**Example & Result:**

```
onworks@onworks:~$ unzip Shivam.zip
Archive: Shivam.zip
replace Shivam.txt? [y]es, [n]o, [A]ll, [N]one, [r]ename: n
onworks@onworks:~$
```

- DevOps Tool Commands

34. Command: git**Purpose:** Version control**Example & Result:**

```
student@Riya:~/Radha$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/student/Radha/.git/
student@Riya:~/Radha$
```

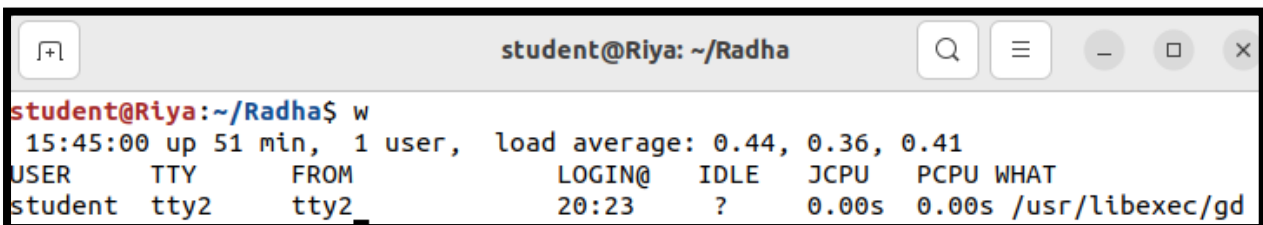
35. Command: docker**Purpose:** Containerization**Example & Result:**

```
onworks@onworks:~/student$ docker run -d -p 8080:80 nginx
Command 'docker' not found, but can be installed with:
sudo snap install docker           # version 20.10.24, or
sudo apt install podman-docker     # version 3.4.4+ds1-1ubuntu1.22.04.2
sudo apt install docker.io        # version 20.10.25-0ubuntu1~22.04.1
See 'snap info docker' for additional versions.
```

36. Command: jenkins**Purpose:** CI service**Example & Result:**

```
onworks@onworks:~/student$ sudo systemctl start jenkins
[sudo] password for onworks:
sudo: a password is required
onworks@onworks:~/student$
```

- **Extra Commands**

37. Command: w**Purpose:** Displays logged-in users.**Example & Result:**

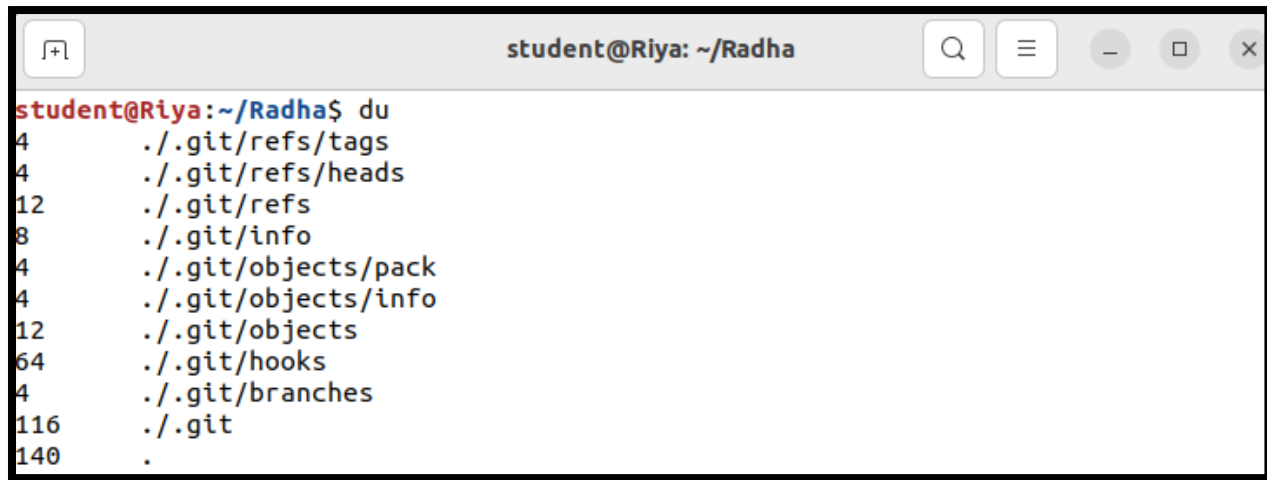
```
student@Riya: ~/Radha$ w
15:45:00 up 51 min,  1 user,  load average: 0.44, 0.36, 0.41
USER      TTY      FROM            LOGIN@   IDLE   JCPU   PCPU WHAT
student   tty2     tty2            20:23    ?      0.00s   0.00s /usr/libexec/gd
```

38. Command: history**Purpose:** Shows command history.**Example & Result:**

```
student@Riya:~/Radha$ history
1073  chown althia:group abc.txt
1074  chown username abc.txt
1075  gedit a.txt
1076  gedit ab.txt
1077  sed -i '3s/director/manager/' ab.txt
1078  sed -i '3s/director/manager/' ab.txt
1079  cat ab.txt
1080  gedit ab.txt
1081  sed -e '3s/director/manager/' ab.txt
1082  gedit ab.txt
1083  sed -i '3s/director/manager/' ab.txt
1084  ls -l
1085  free
1086  sed -i '3s/director/manager/' ab.txt
1087  cat ab.txt
1088  sed -i '3s/Director/Manager/' ab.txt
1089  cat ab.txt
1090  gedit file001.txt
1091  sed '3q' file001.txt
```

39. Command: du

Purpose: Shows directory size.

Example & Result:

```
student@Riya: ~/Radha
student@Riya:~/Radha$ du
4      ./git/refs/tags
4      ./git/refs/heads
12     ./git/refs
8      ./git/info
4      ./git/objects/pack
4      ./git/objects/info
12     ./git/objects
64     ./git/hooks
4      ./git/branches
116    ./git
140    .
```

40. Command: date

Purpose: Displays date and time.

Example & Result:

```
student@Riya:~/Radha$ date
Tuesday 13 January 2026 03:44:16 PM IST
```